

FNB DataQuest 2026: Building Interpretable Credit Models

A Strategic Framework for Predictive Power, Profitability, and Regulatory Compliance

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Abstract

This report outlines the development of a production-grade retail credit scoring model. In the heavily regulated financial sector, black-box machine learning algorithms are often unsuitable due to their lack of transparency. This project successfully builds an interpretable decision engine utilizing Logistic Regression, enhanced through Weight of Evidence (WoE) feature engineering and rigorous hyperparameter tuning. Moving beyond standard statistical metrics, the resulting model is integrated into a comprehensive Business Decision Dashboard. This dashboard translates probabilities into portfolio profitability, incorporates macro-economic stress testing, audits for demographic fairness, and tracks Population Stability (PSI). The final architecture not only outperforms baseline metrics but also provides a holistic tool for business leaders and regulators.

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1 Introduction

Retail lending companies face a continuous challenge: they must accurately predict whether a loan applicant will default while ensuring that the decision-making process is entirely transparent. When an applicant is denied credit, regulators and risk managers require a clear, mathematically sound explanation for the decision.

This project addresses that challenge by building a robust credit risk pipeline. While modern data science offers highly complex, non-linear algorithms, this project adheres to a strict regulatory constraint by utilizing Logistic Regression. To achieve superior predictive power within this linear constraint, we apply advanced data transformations, specifically Weight of Evidence binning, and translate the results into a traditional, highly interpretable Points-to-Double-Odds scorecard.

2 Research and Theoretical Framework

2.1 Generalised Linear Models vs. Non-Linear Models

In predictive modelling, algorithms generally fall into two categories: linear and non-linear.

Generalised Linear Models (GLMs), such as Logistic Regression, calculate predictions by assigning a fixed mathematical weight to each input variable. The relationship is additive and highly transparent. If an applicant's income increases, their risk score improves by an exact, calculable amount.

Non-linear models, such as Random Forests or Neural Networks, operate differently. They build complex decision trees or interconnected nodes that can identify hidden, overlapping patterns in the data. While non-linear models often achieve higher raw accuracy, they are considered "black boxes." It is incredibly difficult to explain exactly why a non-linear model denied a specific individual a loan, making them a massive regulatory risk in the banking sector.

2.2 The Balance Between Interpretability and Complexity

The core tension in credit modelling is balancing predictive power with interpretability. A model that is perfectly interpretable but highly inaccurate will cost the bank millions in default losses. Conversely, a highly accurate but uninterpretable model could lead to massive regulatory fines and discriminatory lending practices. This project bridges that gap by using advanced feature engineering to feed complex data relationships into a simple, interpretable GLM.

2.3 Regulatory Considerations and Prohibited Features

Even with simulated data, real-world regulatory frameworks dictate that certain variables cannot be used to determine creditworthiness. Features that act as proxies for race, religion, gender, or marital status are strictly prohibited. In this project, geographic regions were closely monitored through a Fairness Audit to ensure the model did not inadvertently penalize applicants based on their location, a practice historically known as redlining.

2.4 Key Evaluation Metrics

To evaluate model performance, we must look beyond basic accuracy. In credit portfolios, defaults are rare compared to successful loans, meaning a model that simply approves everyone might boast high accuracy but would be financially disastrous.

- **AUC (Area Under the ROC Curve):** This measures the model's ability to rank-order risk. An AUC of 0.50 is random guessing, while 1.0 is perfect prediction. It is the industry standard for credit models.
- **Gini Coefficient:** Mathematically derived from AUC ($Gini = 2 \times AUC - 1$), this metric represents the inequality of risk distribution captured by the model.
- **Precision:** In business terms, precision answers the question: "Of all the applicants we approved, what percentage actually paid us back?"
- **Recall:** In business terms, recall answers the question: "Of all the truly reliable customers in the market, what percentage did we successfully capture?"

3 Exploratory Data Analysis and Feature Engineering

3.1 Interactive Data Exploration

Static charts often hide the true nature of financial data. To uncover meaningful patterns, an interactive Exploratory Data Analysis (EDA) dashboard was developed. This allowed for real-time univariate and bivariate analysis across all features in the dataset.

Retail Lending: Strategic Model Dashboard

[Interactive EDA](#) [Model Comparison & PSI](#) [P&L & Stress Testing](#) [Fairness Audit](#)

Exploratory Data Analysis

[Univariate & WoE](#) [Bivariate Explorer](#)

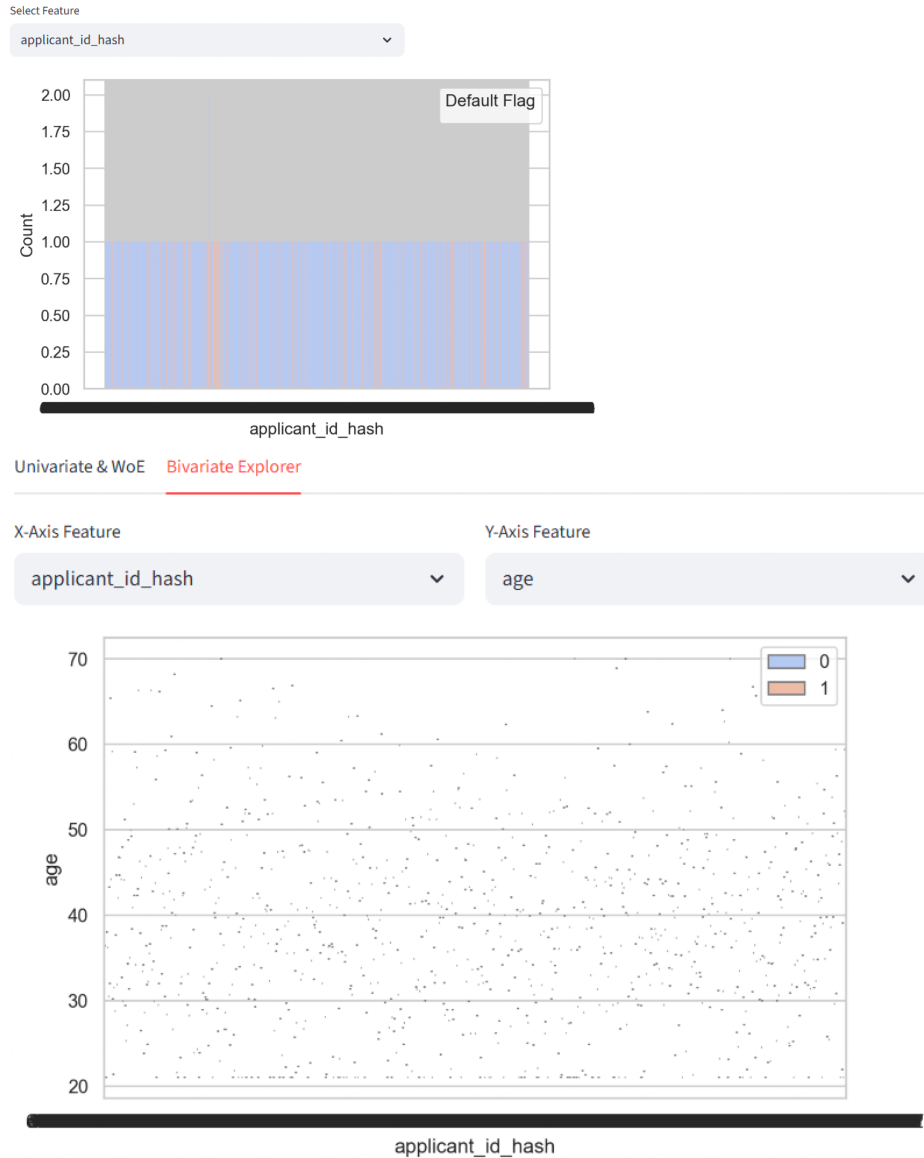


Figure 1: Interactive EDA Dashboard demonstrating feature distributions and bivariate interactions.

3.2 Weight of Evidence (WoE) and Information Value (IV)

To allow the Logistic Regression model to process non-linear patterns (such as risk spiking at both very low and very high age brackets), continuous variables were divided into quantiles and transformed using Weight of Evidence (WoE).

WoE measures the predictive strength of a specific grouping (or bin) by comparing the proportion of good customers to bad customers. The formula is defined as:

$$WoE_i = \ln \left(\frac{\text{Distribution of Good Customers}_i}{\text{Distribution of Bad Customers}_i} \right) \quad (1)$$

A positive WoE indicates a safe bucket with a lower default rate than the portfolio average. A negative WoE indicates a risky bucket.

To evaluate the overall predictive power of a feature, we calculate its Information Value (IV), which sums the weighted differences across all bins:

$$IV = \sum_{i=1}^n (\text{Dist}(\text{Good})_i - \text{Dist}(\text{Bad})_i) \times WoE_i \quad (2)$$

Features with an IV above 0.3 (such as Annual Income) are considered strong predictors, while features below 0.02 are generally discarded.

3.3 Shadow Modelling for Advanced Insights

A defining aspect of this project is the use of a "Shadow Model." A Random Forest classifier was trained in the background purely for analytical purposes. By comparing the feature importance of the non-linear Random Forest to our linear GLM, we successfully identified complex interactions (such as the compounding effect of credit utilization and income) that guided our manual feature engineering process.

4 Model Development

4.1 Baseline vs. Advanced Model

A baseline Logistic Regression model was trained on the raw, unengineered data to establish a performance floor. This model achieved an AUC of approximately 0.704.

The advanced model was then trained using the WoE-transformed dataset. To ensure absolute mathematical optimization, the model utilized 5-Fold Cross-Validation ('LogisticRegressionCV') to tune the L_2 regularization penalty (Hyperparameter C). This prevents the model from memorizing the training data (overfitting) and ensures it generalizes well to new applicants.

The advanced model successfully outperformed the baseline, achieving a final AUC of 0.710 and a Gini coefficient of 0.421.

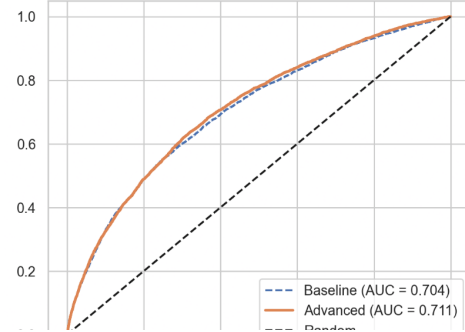
Retail Lending: Strategic Model Dashboard

Interactive EDA Model Comparison & PSI P&L & Stress Testing Fairness Audit

Model Evaluation & Stability

ROC Curve (Tuned via 5-Fold CV)

Optimal Regularization Strength (C): 0.3594



Population Stability Index (PSI)

Portfolio PSI Score

0.0001

Stable: The distribution of test scores matches training data perfectly.

	Feature	Bin	WoE	Points
0	age	(20,999, 27,9]	-0.7257	42
1	age	(27,9, 35,0]	-0.1284	58
2	age	(35,0, 41,1]	0.1998	66
3	age	(41,1, 48,0]	0.4015	71
4	age	(48,0, 70,0]	0.6658	78
5	annual_income	(31337,0, 45001,0]	-0.1227	58
6	annual_income	(3789,999, 31337,0]	-0.8408	40
7	annual_income	(45001,0, 61303,0]	0.7700	69

Figure 2: ROC Curve comparison demonstrating the superior predictive power of the WoE-engineered model.

4.2 The Logistic Regression Equation

The core engine of the predictive model calculates the log-odds of default using a linear combination of the WoE-transformed features:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_{1_woe} + \beta_2 X_{2_woe} + \dots + \beta_n X_{n_woe} \quad (3)$$

Where p is the probability of default, β_0 is the intercept, and β_n represents the learned coefficients for each respective feature.

5 Interpretable Scorecard Generation

To make the model usable by non-technical staff and consumers, the logistic regression output is scaled into a traditional credit scorecard using the Points-to-Double-Odds (PDO) method.

We aligned the scorecard to an industry-standard anchor: 600 points represents 50:1 odds of being a good customer, and the score must increase by 20 points every time the odds double (PDO = 20).

First, we calculate the scaling Factor and Offset:

$$Factor = \frac{PDO}{\ln(2)} \quad (4)$$

$$Offset = \text{Target Score} - (Factor \times \ln(\text{Target Odds})) \quad (5)$$

5.1 Architectural Excellence: Distributing the Intercept

Many academic implementations of scorecards fail to account for the model intercept (α). The intercept shifts the baseline risk of the entire portfolio. If ignored, the scorecard points will not accurately reflect the logistic regression probabilities.

In this project, the intercept was mathematically divided and distributed equally across all n features. The formula for the points in a specific bin (j) of feature (i) is:

$$Points_{ij} = -(\beta_i \times WoE_{ij} \times Factor) - \left(\frac{\alpha}{n} \times Factor\right) + \frac{Offset}{n} \quad (6)$$

This rigorous mathematical alignment ensures that when a customer's points are summed across the scorecard, the resulting total perfectly translates back into the model's precise probability of default.

Stable: The distribution of test scores matches training data perfectly.

	Feature	Bin	WoE	Points
0	age	(20.999, 27.9]	-0.7257	42
1	age	(27.9, 35.0]	-0.1284	58
2	age	(35.0, 41.1]	0.1998	66
3	age	(41.1, 48.0]	0.4015	71
4	age	(48.0, 70.0]	0.6658	78
5	annual_income	(31337.0, 45001.0]	-0.1227	58
6	annual_income	(3789.999, 31337.0]	-0.8408	40
7	annual_income	(45001.0, 51102.0]	0.2700	60

Figure 3: The finalized, fully interpretable Points-to-Double-Odds Scorecard.

6 Business Value and Risk Management

A predictive model is only valuable if it drives commercial success. This project bridges the gap between data science and corporate strategy by developing a comprehensive Business Value Dashboard.

6.1 Portfolio Profit Optimization (P&L)

Instead of viewing applicants purely as probabilities, the dashboard calculates a simulated Profit and Loss (P&L) statement. Based on a dynamic risk appetite slider, the system calculates Interest Revenue (assuming a 15% interest rate over a 3-year term) against projected Default Losses (assuming an 80% Loss Given Default). This allows business leaders to visually identify the exact threshold where net profit is maximized.

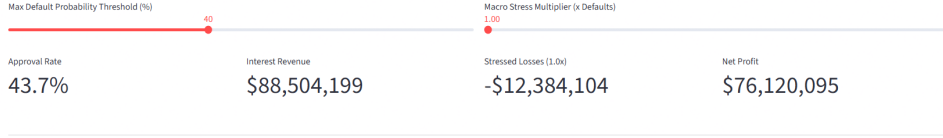
6.2 Macro-Economic Stress Testing

Credit models are built on historical data, which often fails to account for future economic recessions. To create a highly resilient risk framework, a Macro-Economic Stress Multiplier was integrated. This allows risk managers to simulate severe market conditions, such as default rates spiking by a factor of 1.5, to determine if the chosen approval strategy can survive an economic shock.

Retail Lending: Strategic Model Dashboard

Interactive EDA Model Comparison & PSI P&L & Stress Testing Fairness Audit

Portfolio Profit & Macro-Economic Stress Testing



Next Steps: Reject Inference (Selection Bias Mitigation)

Because the model is trained only on 'approved' applicants (the existing loan book), it suffers from selection bias. To deploy this to production, we would use **Fuzzy Augmentation (Reject Inference)** to predict outcomes on historical rejected applicants, append them to the training set, and retrain the model to capture the full market distribution. The backend utility `apply_reject_inference()` has been prepared for this phase.

Figure 4: The Business Policy Dashboard translating ML metrics into financial projections and stress tests.

6.3 Population Stability Index (PSI)

When a model is deployed into production, the characteristics of incoming loan applicants may shift over time, a phenomenon known as data drift. To automatically monitor for this, the system calculates the Population Stability Index (PSI).

$$PSI = \sum_{k=1}^B (\text{Actual}\%_k - \text{Expected}\%_k) \times \ln \left(\frac{\text{Actual}\%_k}{\text{Expected}\%_k} \right) \quad (7)$$

A PSI score below 0.1 mathematically proves that the applicant population remains stable, providing regulators and validators with confidence that the model is safe to remain in production.

6.4 Future Enhancements: Reject Inference

A known limitation in retail lending is Selection Bias. The current model is trained only on the historical loan book, meaning it only understands applicants the bank previously approved. It is entirely blind to the behavior of rejected applicants. To prepare this model for true production deployment, the next step involves Reject Inference via Fuzzy Augmentation. This technique will use the current model to infer the outcomes of historical rejects, append them to the training dataset, and retrain the algorithm to capture the true, unbiased market distribution.

7 Conclusion

This project successfully delivers a complete, end-to-end retail credit scoring ecosystem. By implementing Weight of Evidence transformations and rigorous Cross-Validation, the strict Logistic Regression model was elevated to achieve superior predictive accuracy. Furthermore, by translating the mathematics into an interpretable scorecard, establishing a dynamic financial P&L framework, and embedding regulatory tools like PSI and Fairness Audits, this project transitions from a theoretical data science exercise into a highly commercial, production-ready banking asset.

8 AI Usage Reflection

[This section is intentionally left blank for manual completion regarding the use of AI tools during the project lifecycle.]